

The Effect of LIHTC on Housing Markets: Quantities, Rents, and Welfare

Fernando Ferreira, Gillian Meyer, and Matthew Turner

September 21, 2026

LIHTC is a developer based subsidy program. There were about 2.8m LIHTC units in 2023. At 150,000\$ each, this is 420b\$ worth of apartments.

We are interested in how the program affects housing markets.

1. Does LIHTC increase the stock of rental housing?
2. Does LIHTC affect rent?
3. Does LIHTC have important unintended or external consequences?
4. How much surplus does LIHTC generate, and who gets it?

We estimate block-group level housing market outcomes using a DiD organized around the block-groups subject to winning and losing LIHTC applications.

We use estimated treatment effects from a LIHTC award/project as the basis for a simple welfare analysis.

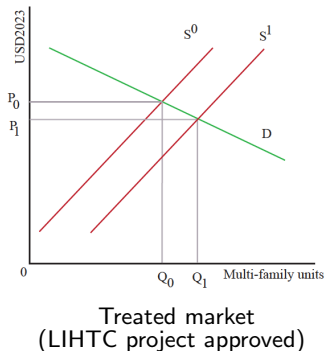
How LIHTC works I

Established by the 1986 Tax Reform Act and run by the IRS

- ▶ Federal government awards LIHTC tax credits to states by formula
- ▶ Developers apply to state housing authorities for LIHTC tax credits with a detailed description of their project
- ▶ State housing authorities award LIHTC credits to applications in an about annual competition using (mostly) federally determined criteria.
- ▶ Winners get about 70% of non-land costs and are subject to a rent cap (18% of AMI) for 30 years and must rent to households with income less than 60% AMI.

Welfare/Model I

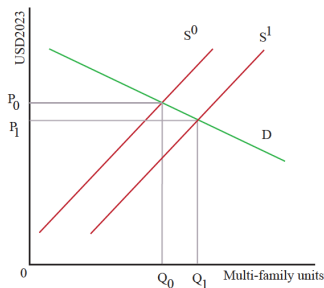
Start with equilibrium (P_0, Q_0) . Approval of a LIHTC project shifts quantity supplied in the treated local market from Q_0 to Q_1 .



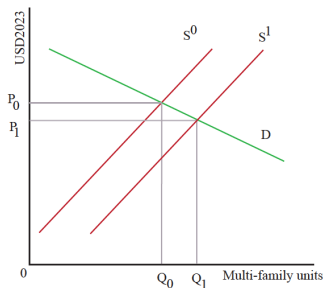
- In this example, LIHTC increases supply and lowers rent.
- We estimate $(\Delta P, \Delta Q)$ and calculate change in surplus and transfers. Identifying restriction is that LIHTC doesn't shift demand.

Welfare/Model II

We compare the treated market with a control market that had a rejected LIHTC project:



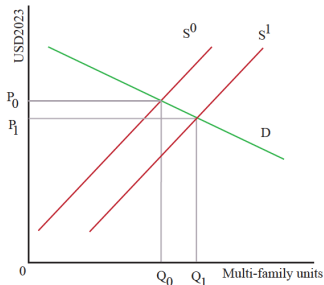
Treated market
(LIHTC project approved)



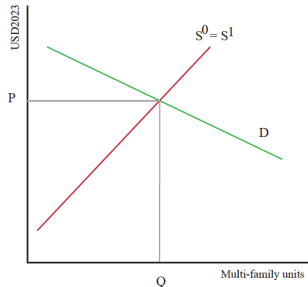
Control market
(LIHTC project rejected)

- In this example, there is **100% crowd-out** no effect on price. Developers in the control market would have constructed the same number of units in the absence of LIHTC tax credits.

Welfare/Model III



Treated market
(LIHTC project approved)



Control market
(LIHTC project rejected)

- In this example, there is about **0 crowd-out**. Developers would *not* have constructed the new multifamily units in the control market in the absence of LIHTC tax credits.

Literature I

- ▶ 100% Crowd out $\implies$ 100 LIHTC units changes housing stock by zero units. 0% Crowd out $\implies$ 100 LIHTC units changes housing stock by 100 units.
 - ▶ Sinai and Waldfogel [2005]. 1990 Cross-section, county and MSA. 50-70% crowd-out for all public housing units.
 - ▶ Eriksen and Rosenthal [2010]. 1990-2000 first differences, MSA, county, 10k disk around a tract. OLS 15% crowd out, IV 100% crowd out for LIHTC.
 - ▶ Baum-Snow and Marion [2009] RDD on threshold for QCTs in 1990-2000. About 20% crowd out of rental units.
 - ▶ Soltas [2024]. Similar win/lose design used in this paper, but compares LIHTC projects at parcel level. 75% crowd out (a different definition).

Literature II

- ▶ Prices:
 - ▶ Baum-Snow and Marion [2009] find 10% increase in price of owner occupied units from 100 LIHTC units in block-group.
 - ▶ Diamond and McQuade [2019], zero average effect on owner occupied, -2%(+6%) in wealthy(poor) neighborhoods, decaying to 0% 1km from LIHTC project. Mcculloch [2025] reanalysis gets -2%.
 - ▶ Burge [2011], LIHTC $\approx$ 10% below hedonic price of market rate rental units in one city. Soltas [2024] LIHTC units 12% below comparable market rate rental units.
- ▶ Quality: Burge [2011] LIHTC units tend to be newer than market units (126 buildings in Tallahassee in 2002).
- ▶ Externalities: Baum-Snow and Marion [2009], decline in tract income, increase in share renters, and faster turnover, but this looks mechanical. Diamond and McQuade [2019] decreases in crime near LIHTC units in poor neighborhoods.

Literature III

Summary:

- ▶ LIHTC is hard to study because (1) outcomes are hard to observe (e.g. demographics and rent of LIHTC residents, locations of LIHTC projects) and, (2) unraveling the assignment process is difficult.
- ▶ From 25 years of research we conclude
 - ▶ Crowd-out is between 20% and 100%. The two most compelling research designs give 20% and 75%.
 - ▶ Prices of owner occupied housing near LIHTC change between -2% and $+10\%$.
 - ▶ LIHTC units are newer on average and rent for about 12% less than comparable market rate units, on average.
- ▶ There has been surprisingly little interest in *market* level rental prices and housing consumption.

Data I

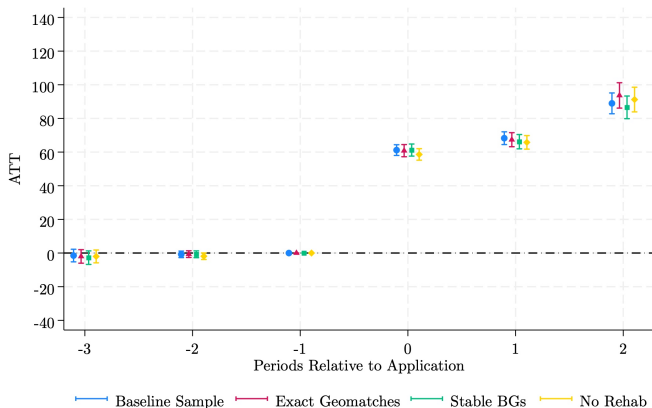
We use three main data sources

- ▶ Purpose collected and digitized lists of applications and awards from State Housing Authority websites and a commercial data aggregator (Application data). Each application reports state, year, successful or not, address, # units, rehab vs. new, cost of units.
- ▶ Housing and Urban Development (HUD) data inventorying all LIHTC awards and units in service. Each award/project has a date, number of units, and geocoded latitude and longitude.
- ▶ Census/American Community Survey block group level data for 1990, 2000, 2005-2009, 2010-2014, 2015-2019, 2019-2024. We standardize to 2010 boundaries. This is where we get most of our outcomes.

Research design I

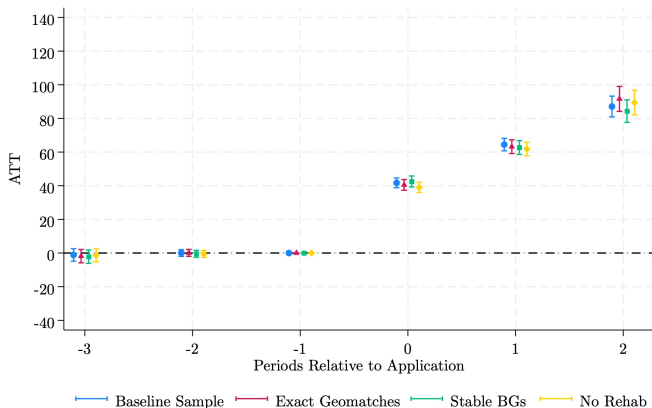
- ▶ Unit of observation is a block-group period.
- ▶ A period is a 5 (10) year ACS (Census) window; 1990, 2000, 2005-2009, 2010-2014, 2015-2019, 2019-2024. Six periods.
- ▶ Staggered adoption DiD. We control for state fixed effects in all estimations. We use the doubly robust estimator [Sant'Anna and Zhao, 2020].

Main sample, HUD units awarded



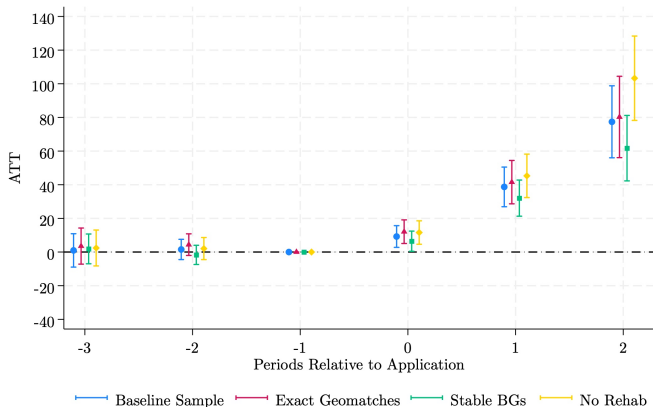
State awards in our application data show up in HUD data. The height is important, ≈ 87 units.

Main sample, HUD units Placed in service.



There is a construction lag between award and placed in service.
2-3 ACS periods post award, all awarded units are built.

Main sample, ACS Multi-family units.



2-3 ACS periods post award, ACS Multifamily converges to HUD units awarded.

Crowd out?

Successful application

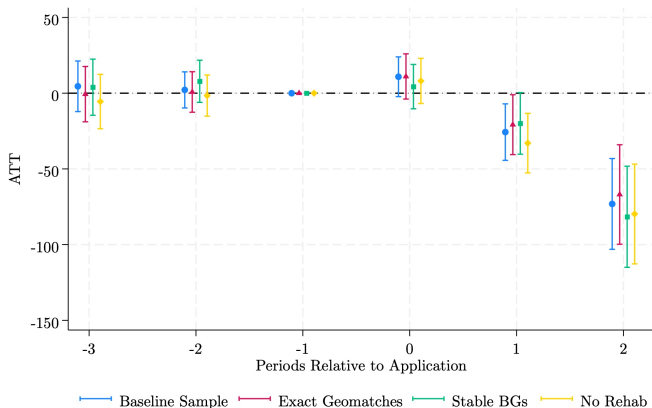
⇒ 89 LIHTC units awarded in HUD data

⇒ 87 placed in service in HUD data

⇒ 77 multi-family units in ACS

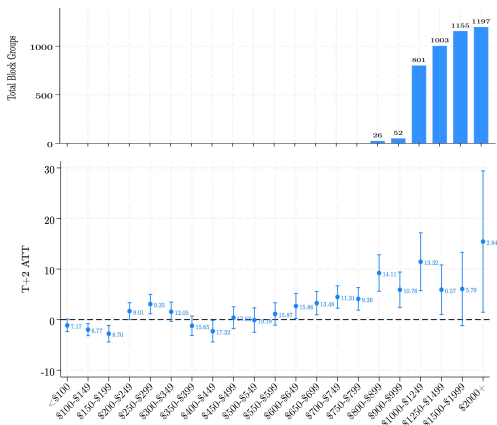
⇒ $\frac{77}{89} = 13\%$ crowd-out. 95% CI $[-6\%, 28\%]$

Main sample, ACS Gross Rent.



Average gross rent declines by about 73\$ month per unit in treated block groups.

Main sample, change in rental units by initial rent.



Top: Histogram of block-groups by LIHTC cap. Bottom: change in total rental units by rent bin in treated block-group. Much of the increase in units comes far from the LIHTC caps. Units for the very poor are not affected.

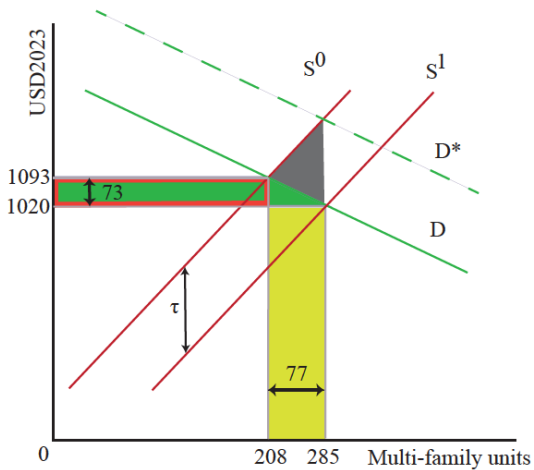
Effect on Demand?

We want to know whether a LIHTC award changes the demand for rental housing in the block-group.

- ▶ No/tiny change to demographics of residents.
- ▶ An average apartment gets bigger (change in average rent is understated).
- ▶ No change in turnover.
- ▶ No change in owner occupied price.

We cannot find anything that seems to indicate that the change in rental price is a consequence of a change in demand.

Welfare I



Welfare II

With supplemental information about the cost to build a LIHTC unit, the size of the subsidy, and an assumed 10% discount rate over a 30 year lifetime, this figure lets us conclude that an average LIHTC subsidy of \$89,000/unit causes (in PV);

- ▶ Increase in consumers surplus of \$26.2 thousand per new unit. 85% is a transfer from landlords.
- ▶ New rental income \$115.4 thousand per unit. Cost of a unit is \$150 thousand.
- ▶ An average unit loses \$35 thousand before the subsidy, has a profit of \$54.7 thousand per unit after the subsidy.
- ▶ Deadweight loss \$31 thousand per unit.
- ▶ Gap between social and private value of housing required for this subsidy to be optimal is 54% of market rent.

Sample average LIHTC cap is \$1125 > \$1093 average market rent.
A binding rent cap shifts rent revenue from developers to tenants.

Welfare III

- ▶ We find some evidence for heterogeneity in price and quantity effect between coastal and Sun Belt areas. This will lead to different conclusions.
- ▶ Our analysis is based on a 7% rent decrease. Our empirical results also support a 2.5% decrease.
- ▶ There is evidence that LIHTC cap binds and rents on LIHTC units are about 12% below market. This would explain much of the drop in average rent.

Conclude

We investigate the effect of LIHTC on housing markets. We find

- ▶ A marginal LIHTC unit awarded adds about one multi-family rental unit to the housing stock. There is essentially zero crowd out.
- ▶ An average LIHTC project reduces rental prices by about \$73/month.
- ▶ On average, LIHTC transfers \$89k per unit from tax payers to developers. This leads to a \$26 thousand transfer to tenants, \$54 thousand of profit for developers, and deadweight loss of \$31 thousand.
- ▶ Rationalizing the average level of LIHTC subsidy requires that the social value of housing exceed market value by about 54%.
- ▶ The subsidy is probably larger than required to elicit developer participation.

Bibliography I

- Todd Sinai and Joel Waldfogel. Do low-income housing subsidies increase the occupied housing stock? *Journal of public Economics*, 89(11-12):2137–2164, 2005.
- Michael D Eriksen and Stuart S Rosenthal. Crowd out effects of place-based subsidized rental housing: New evidence from the lihtc program. *Journal of Public Economics*, 94(11-12):953–966, 2010.
- Nathaniel Baum-Snow and Justin Marion. The effects of low income housing tax credit developments on neighborhoods. *Journal of Public Economics*, 93(5-6):654–666, 2009.
- Evan Soltas. Tax incentives and the supply of low-income housing. *Unpublished manuscript*, 2024.

Bibliography II

Rebecca Diamond and Tim McQuade. Who wants affordable housing in their backyard? an equilibrium analysis of low-income property development. *Journal of Political Economy*, 127(3): 1063–1117, 2019.

Sean Mcculloch. Continuous spatial difference in differences for welfare evaluation. *Unpublished manuscript*, 2025.

Gregory S Burge. Do tenants capture the benefits from the low-income housing tax credit program? *Real Estate Economics*, 39(1):71–96, 2011.

Pedro HC Sant’Anna and Jun Zhao. Doubly robust difference-in-differences estimators. *Journal of econometrics*, 219(1):101–122, 2020.